EXTREMAL GRAPH THEORY

Erdős Problem 915

From Proof to Search

One extremal question, carried across twelve variants.

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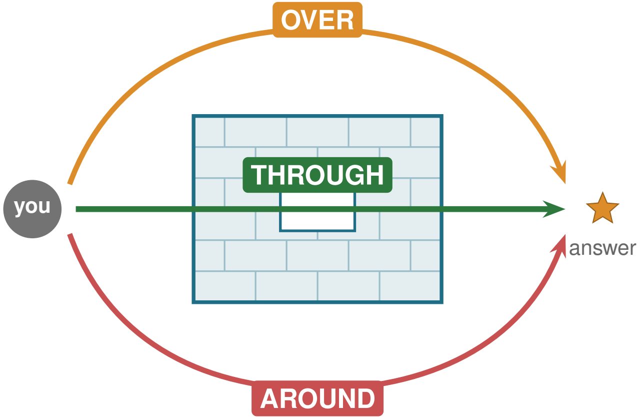
A hard problem is a wall



Facing an obstacle, there are three ways past it. Mathematics has the same three.

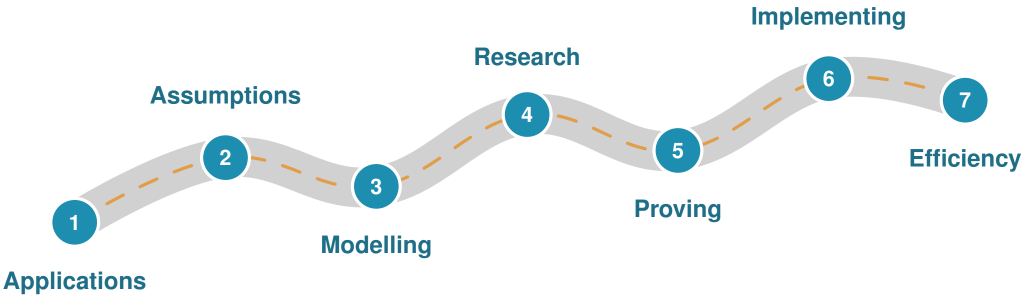
Three ways past the wall

The "best solution" of an extremal problem is the densest graph allowed by the rule. Three routes reach it.

THROUGH	OVER	AROUND
Proof	**Calculation**	**Approximation**
A single clean argument that settles every n at once.	Check a finite domain exactly, by machine, and emit a certificate.	Search for good examples. Gives lower bounds, never the front door.
Mader, Menger, Gomory-Hu, a hand induction.	Exact max-flow, cut-counting upper bounds.	Simulated annealing, tabu search.

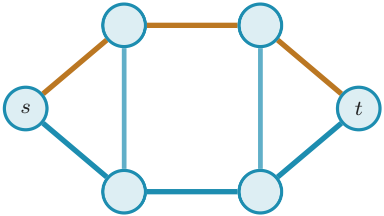
This thesis uses all three, and is honest about which one each answer rests on.

The road of a research problem



Every problem travels this road. The slides ahead follow it, from the rule to the running code.

Problem 915, precisely



two edge-disjoint s-t routes

A graph G: n vertices, E edges.

Independent routes. $\lambda_G(u,v)$ is the largest number of edge-disjoint u-v paths. By Menger, also the cheapest cut between them.

Binding connectivity.

$$\lambda_G^{\max} = \max_{u,v} \lambda_G(u,v).$$

The rule. No pair gets m routes: $\lambda_G^{\max} \le m-1$.

The goal. Pack in as many edges as the rule allows:

$$\ell_m(n) = \max\{\, |E| : \lambda_G^{\max} \le m-1 \,\}.$$

cap connectivity at $m-1 \implies \ell_m(n) = \left\lfloor \frac{m(n-1)}{2} \right\rfloor$ (Mader)

What is it good for?

Honest answer: no direct application. But in mathematics the road often matters more than the destination.

The question

How do you quantify the redundancy of a network: the spare, independent routes it keeps between its points? Graphs turn that into a vast space to search.

The transferable tools

- maximum-flow connectivity
- temperature-guided search
- machine-checked certificates

Where redundancy lives

- electricity grids
- pipelines
- road and rail networks

Backup routes everywhere.

The methods outlive the single number they were built to find.

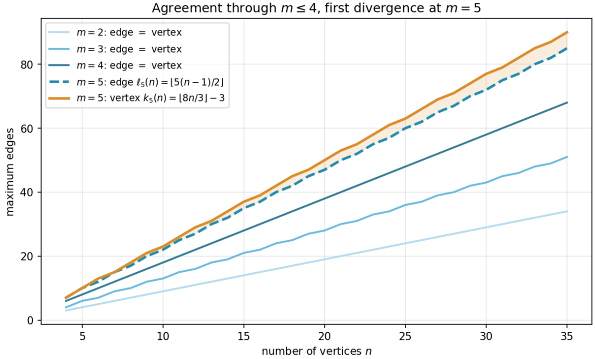
One question, twelve variants

Change the rules of the network one axis at a time.

Model	simple	multigraph	hypergraph	$\times 3$	
Direction	undirected	directed		$\times 2$	$= 12$
Separation	edge-disjoint	vertex-disjoint		$\times 2$	

Mader's clean theorem is one corner. The other eleven are the journey, and they get harder as the model gets richer.

Edge routes versus vertex routes



Two separations: routes may avoid sharing an **edge** (λ) or, more strictly, an internal **vertex** (κ). Whitney: $\kappa \leq \lambda$.

$m \leq 4$ the two problems give the **same** answer (Leonard).

$m = 5$ first divergence (Sørensen-Thomassen):

$$k_5(n) = \left\lfloor \frac{8n}{3} \right\rfloor - 3 \;>\; \ell_5(n).$$

$m \geq 6$ open since 1974. The extremal structure is genuinely not understood. The one direction the thesis does not try to force.

What fell, and how hard

How each variant is known, from a clean argument to a stubborn blank.

easier → harder

Proved by hand
- undirected edge (simple, multi, hyper)
- multigraph vertex
- directed $m = 2$

Certified by machine
- directed multigraph value, $n \le 6$
- hypergraph vertex $m \le 3$

Conjecture + lower bound
- directed arc, $m \ge 3$
- directed vertex, $m \ge 3$
- directed multigraph $m = 3$ (conditional)

Open
- vertex-disjoint $m \ge 6$
- hypergraph vertex $m \ge 4$
- directed hypergraph

The honesty contract: search proves nothing on its own. A value is “settled” only with a matching hand proof or a machine certificate.

How the machine works

Search engine: annealing vs tabu

Both walk the space of graphs. Equal 6 s budget.

variant	opt.	SA	tabu
simple undirected $n=7$	9	9	$\mathbf{9}$
simple directed $n=7$	18	16	$\mathbf{18}$
directed multi $n=7$	24	20	$\mathbf{24}$

Tabu's best-improving step climbs the density gradient and reaches the directed optima where annealing plateaus.

Implementation: Python with a C hot path

Python core: `numpy` + `scipy` max-flow (every connectivity is one integer flow)

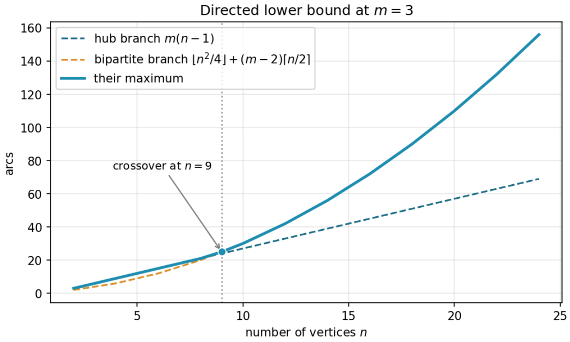
optional `_erdos_fast.so` hot path (same answers, pure-Python fallback)

Measured speedups:

- dense max-flow $\approx 2\times$
- connectivity test $2.2\times$
- canonical form $\mathbf{147}\times$

Annealing stays the default (its restarts parallelise trivially); tabu is the sharper tool past the certified range.

Results on the directed frontier



Two competing constructions at $m = 3$. The linear hub leads early, the quadratic augmented-bipartite family overtakes near $n = 9$.

Two genuine advances:

solved **Hypergraph vertex problem** settled for all r and all n at $m \leq 3$:

$$k_m^{(r)}(n) = \left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor,$$

via a new rank bound on incidence graphs.

reduced **Directed multigraph, $m = 3$** the whole infinite family collapses to **two finite checks** on 7 vertices ($L_3^{\mathrm{dir}}(7) = 24$ and its extremisers).

The whole landscape

Separation	Model	Value	Status
Undirected			
edge	simple	$\lfloor m(n-1)/2 \rfloor$	proved (Mader)
edge	multi	$(m-1)(n-1)$	proved
edge	hyper	$\lfloor (m-1)(n-1)/(r-1) \rfloor$	proved (Gomory-Hu)
vertex	simple	$\lfloor m(n-1)/2 \rfloor$ $(m \leq 4)$, $\lfloor 8n/3 \rfloor - 3$ $(m{=}5)$	cited, open $m \geq 6$
vertex	multi	parallel edges do not help	proved
vertex	hyper	$\lfloor (m-1)(n-1)/(r-1) \rfloor$ $(m \leq 3)$	proved $m \leq 3$, open $m \geq 4$
Directed			
arc	simple	$\max(2(n-1), \lfloor n^2/4 \rfloor)$ $(m{=}2)$	proved, conj. $m \geq 3$
arc	multi	$(m-1)\max(2(n-1), \lfloor n^2/4 \rfloor)$	proved $n \leq 6$, cond. $m{=}3$
arc	hyper	quadratic in n, exact value open	lower bound
vertex	simple	$\max(2(n-1), \lfloor n^2/4 \rfloor)$ $(m{=}2)$	proved, conj. $m \geq 3$
vertex	multi	reduces to simple digraph	proved
vertex	hyper	quadratic in n, exact value open	lower bound

proved **certified** **conjecture** **open**

What is left, and why it matters

Open problems

- **Backward-arc lemma.** Proving an extremal non-hub digraph has no arc from B to A turns the directed conjecture into a theorem for all $m \geq 3$.
- **Two finite $n = 7$ checks** close the directed multigraph problem at $m = 3$. Memory-safe today, runtime-limited. Gurobi or the cluster would finish them.
- **Vertex-disjoint** $m \geq 6$, open since 1974.
- **Hypergraph vertex** $m \geq 4$, needing a 4-connectivity decomposition.

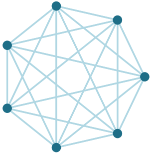
The contribution

- one representation for all twelve variants
- one exact connectivity checker
- one certifier for small-case upper bounds
- one cooled search for constructions

The use of building the microscope is not that it answers every question, but that it shows exactly which questions are left.

Thank you

Questions?



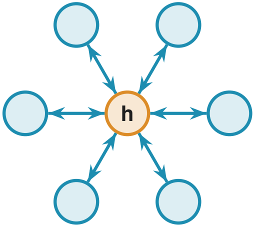
Backup slides follow: a deep dive into each variant.

BACKUP SLIDES

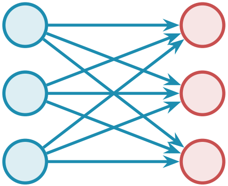
Dive deeper into any single variant.

Backup: the directed arc conjecture

At $m = 2$ the value is proved: $\ell_2^{\mathrm{dir}}(n) = \max(2(n-1), \lfloor n^2/4 \rfloor)$, by an induction deleting a minimum-degree vertex. The linear branch is the bidirected hub, the quadratic branch the one-directional bipartite digraph, tying at $n = 7$.



bidirected hub: $2(n-1)$

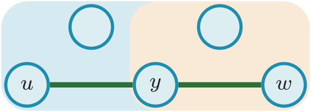


bipartite digraph: $\lfloor n^2/4 \rfloor$

For $m \geq 3$ the conjecture adds the augmented-bipartite family. The missing piece is the **backward-arc lemma** for the matching upper bound.

Backup: the hypergraph vertex problem

Routes in an r-uniform hypergraph are **Berge paths**, alternating vertices and hyperedges. A single hyperedge serves only one route, so two Berge routes are disjoint when they share no hyperedge.



one Berge path u, e_1, y, e_2, w

Result. For $m \le 3$, every uniformity r, all n:

$$k_m^{(r)}(n) = \left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor = \ell_m^{(r)}(n).$$

The $m = 3$ proof rests on a **rank bound for incidence graphs**, derived through Tutte's triconnected (SPQR) decomposition, valid exactly where $\kappa \le 2$.

Open at $m \ge 4$: it would need a 4-connectivity analogue of that decomposition. No early counterexample: $k_4^{(3)}(5) = 6$ machine-checked.

Backup: collapsing an infinite family to two checks

The directed multigraph problem at $m = 3$ is reduced, not assumed, to a finite computation.

Attachment lemma (unconditional) → Odd-step theorem (conditional) → Single induction over all $n \geq 8$ → Two checks on $n = 7$

The two remaining facts, on multigraphs with multiplicities in $\{0, 1, 2\}$:

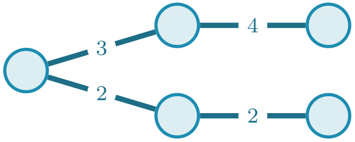
(a) $L_3^{\mathrm{dir}}(7) = 24$;

(b) the only 24-arc extremisers (max degree ≤ 8) are $2B_{3,4}$, $2B_{4,3}$, and the doubled bidirected path P_7.

This bypasses the unproved minimum-degree conjecture entirely. The wall is now runtime, not structure.

Backup: how connectivity is stored

A **Gomory-Hu tree** is a weighted tree on the same vertices in which $\lambda_G(u,v)$ is the lightest edge on the tree path from u to v. All $\binom{n}{2}$ connectivities live in $n-1$ numbers, and $\lambda_G^{\max}$ is simply the heaviest tree edge.



This is exactly the structure the checker and the Mader proof both lean on: store the whole connectivity profile cheaply, read off the binding pair instantly.

Backup: values at a glance

Object	Extremal value
simple undirected edge, $\ell_m(n)$	$\lfloor m(n-1)/2 \rfloor$
multigraph edge, $L_m(n)$	$(m-1)(n-1)$
hypergraph edge, $\ell_m^{(r)}(n)$	$\lfloor (m-1)(n-1)/(r-1) \rfloor$
simple vertex, $k_m(n)$ $(m \le 4)$	$\lfloor m(n-1)/2 \rfloor$
simple vertex, $k_5(n)$	$\lfloor 8n/3 \rfloor - 3$
directed arc, $m = 2$	$\max(2(n-1), \lfloor n^2/4 \rfloor)$
directed multigraph arc	$(m-1)\max(2(n-1), \lfloor n^2/4 \rfloor)$
hypergraph vertex, $m \le 3$	$\lfloor (m-1)(n-1)/(r-1) \rfloor$
random-graph threshold	$p^* = m/n$

n vertices, threshold m (no pair gets m routes), uniformity r.